STILLSTACK · DESIGN DECISION RECORD

StillStack concept selection

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/decisions/0004-concept-selection.md @ 4e01efb		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Option A cost USD 460.70 (panel label, D-05); the selection itself is still proposed, awaiting Amish (no pick)
0.1	2026-10-24	Amish Chadha	Concept selection between option A (four-stage wick still) and option B (single-effect tilted wick still); within 10 %, both to Amish; proposed, awaiting Amish

StillStack concept selection

Status: Proposed, awaiting Amish (SSK-DEC-001 D-03). No option is picked, because the two score within 10 % (STANDARDS section 21).

Context

StillStack was designed before kit 1.8, so its current design is option A, the four-stage wick still, and the repo is not redesigned without Amish's go. The kit 1.9.1 upgrade (2026-10-24) adds option B, a single-effect tilted wick still, with a different working principle: one evaporation and one condensation, the latent heat lost through the cover, rather than four stages that reuse it. Both are modelled to detail level D3 in `cad/src/model.py` and checked in SSK-STM-001. The comparison and scores are in SSK-OPT-001.

Options

1. **Option A, four-stage wick still (current design).** About 10.3 L/m2/day on the design day, USD 460.70 per 1 m2 panel, 18.8 kg dry. Complex: five shaped plates, twenty wick strips, forty clips, a certified food-contact coating; PR-002 not met and several material confirmations open.
2. **Option B, single-effect tilted wick still.** About 3.8 L/m2/day (an assumed 45 % efficiency, near the 4.5 to 5.0 L reported for tilted wick stills), so about 2.6 panels for a household's 10 L. Simpler, glass condensing surface with no coating to certify, one wick, easy to clean; heavier per panel and costlier per litre.
3. **Option A with option B's best idea.** B's glass cover is the one feature worth grafting: a glass outer sheet in place of twin-wall polycarbonate would remove the 120 °C glazing limit (STEM S13) at a mass cost of about 9 kg. It is not proposed now, because it would push the panel over the 20 kg of PR-009.

Scores

Option A scores 152 and option B 153 of 220 (SSK-OPT-001, Table 2): a difference of under 1 %. A wins on the core needs, water per panel and per square metre (UR-001, UR-002); B wins on simplicity, safe materials, cleaning and buildability (UR-003, UR-004, UR-006, UR-007, UR-010, buildability, maintenance).

Recommendation

None, by rule. For Amish's choice: if the measure is water per household for the money and the effort of building, A is ahead (about USD 461 and one panel for 10 L a day, against about 2.6 panels and roughly USD 590 for B); if the measure is the lowest risk to drinking-water safety and the easiest care by a part-time caretaker, B is ahead. The repo stays on option A until Amish decides.

Consequences

- If A: no change; the open items of SSK-DEC-001 (D-04 to D-10) continue.
- If B: a new constructable design, drawings, BOM and build plan for option B at TRL 3, in a later session.